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BAGONG PILIPINAS

CC 101 – Introduction to Computing

"From Problem to README": Designing a Number-System Converter and Documenting It in Markdown

Name: Roxanne E. Gracela	**Block:** BSCS1-1
Computer Unit Used (Desktop/Laptop, brand): Laptop	**Date:** October 16, 2026

I. Objectives

At the end of this activity, you should be able to:

- Analyze a number-system conversion problem by identifying its inputs, processes, loop condition, and outputs.
- Write pseudocode using sequence, selection, and iteration.
- Draw flowcharts using standard symbols and check them by tracing and testing with sample data.
- Create a properly formatted README.md file in Markdown using Notepad, without an Internet connection.
- Upload the README.md to your CC 101 GitHub Computing Portfolio repository (take-home part).

II. Materials Needed

- A computer running Windows 11 with Notepad (no network or Internet connection is needed in the laboratory)
- This worksheet, a pencil, and a ruler (or a flowchart template) for drawing flowcharts
- A USB flash drive, to bring your README.md home for uploading
- For Part G only (outside the laboratory): any device with Internet access and a free GitHub account
- Week 7 lecture notes, for reference on pseudocode, flowchart symbols, and Markdown syntax

III. Background

Programmers do not start by typing code. They first analyze the problem, plan the solution as pseudocode and a flowchart, and check it by hand. In this activity, you will design a program that converts numbers between the decimal and binary number systems you learned in Week 5. An algorithm can be written as pseudocode (English-like steps) or drawn as a flowchart (standard symbols connected by arrows). Good work is also documented. On GitHub, the standard way to introduce a project is a README.md file written in Markdown, a plain-text format that uses symbols such as #, *, and - for headings, emphasis, and lists. Because Markdown is plain text, you can write it in Notepad with no Internet connection and upload it to GitHub later.

IV. Procedure

Part A – Analyze the Problem (Problem 1)

Problem 1: Decimal-to-Binary Converter. Design a program that asks the user for a whole number in **decimal** (base 10) that is 0 or greater, converts it to **binary** (base 2), and displays the binary result.

Worked example (13 in decimal): 13 ÷ 2 = 6 r 1; 6 ÷ 2 = 3 r 0; 3 ÷ 2 = 1 r 1; 1 ÷ 2 = 0 r 1. Reading the remainders from bottom to top gives **1101**.

1. Convert 13 and 25 to binary by hand using the method above, so you understand every step before writing the pseudocode.
2. Complete **Table 1**: list the input, the output, the processes (operations), the decision or loop

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condition, and the variables you will need.

Part B – Write the Pseudocode

3. In **Box 1**, write the pseudocode for Problem 1. Use START, INPUT, SET, WHILE … DO … ENDWHILE, IF … THEN … ELSE … ENDIF, DISPLAY, and END, and indent the steps inside the loop.

   Tip: Use **MOD** for the remainder of a division (13 MOD 2 = 1) and **DIV** for whole-number division (13 DIV 2 = 6). Each new remainder must be placed at the **left** of the binary digits found so far, for example SET binary = remainder + binary. Also decide what your algorithm should display when the input is 0.

Part C – Draw the Flowchart and Desk Check

4. In **Figure 1**, draw the flowchart for Problem 1 using the standard symbols: oval (start/end), parallelogram (input/output), rectangle (process), and diamond (decision). Show the loop with an arrow going back to the decision, and label every decision exit **Yes** or **No**.
5. Trace your pseudocode with the input **13** and fill in one row of **Table 2** for each pass through the loop. The last column should end with 1101.
6. Test your pseudocode and flowchart with the five inputs in **Table 3**. First convert each number by hand, then follow your pseudocode or flowchart step by step. If the two results do not match, find and correct the error in your pseudocode and flowchart.

Part D – Reverse Conversion (Problem 2)

Problem 2: Binary-to-Decimal Converter. Design a program that asks the user for a binary number (entered as digits 0 and 1, such as 1101) and displays its decimal value. Use the **positional (place value) method** from Week 5: starting from the rightmost digit, multiply each digit by its place value (1, 2, 4, 8, 16, …) and add the products.

Worked example (1101 in binary): (1 × 1) + (0 × 2) + (1 × 4) + (1 × 8) = 1 + 0 + 4 + 8 = **13**.

7. In **Box 2**, write the pseudocode. You can take the rightmost digit with **binary MOD 10** and remove it with **binary DIV 10**; the place value starts at 1 and doubles after each digit.
8. In **Figure 2**, draw its flowchart.
9. Test your pseudocode with the inputs in **Table 4**, comparing hand computation with the result of following your pseudocode.
10. **Optional / Bonus:** Extend Problem 1 so the user can choose the target base (2, 8, or 16). For base 16, remainders 10 to 15 must be displayed as the letters A to F. Write the extended pseudocode in the Bonus box.

Part E – Create Your README.md Offline

11. Show file extensions so that you can see the real file name. Press **Windows + E** to open File Explorer, then click **View** → **Show** → **File name extensions** (make sure it is checked).
12. In File Explorer, click **Documents**, then click **New** → **Folder**. Name the folder **cc101-portfolio** (all lowercase, with a hyphen) and press **Enter**. Open it and create two more folders inside: **week07-algorithms** and **images**.
13. Press the **Windows key**, type **Notepad**, and press **Enter**. If your Notepad shows a formatting toolbar, open the **View** menu and switch to the Markdown **syntax** view so the symbols you type stay visible.
14. Type the README content below. Replace everything in square brackets with your own information, and replace the pseudocode with your pseudocode from Box 1. Keep the blank lines; they separate the parts.

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```
# CC 101 Computing Portfolio

This repository contains my laboratory outputs and documentation for *CC 101 Introduction to
Computing* at **Sorsogon State University – Bulan Campus**.

## About Me

- **Name:** [Your full name]
- **Program and Section:** BSCS 1-[section]
- **Interests:** [two or three interests]
- **Goal:** [one sentence about your goal as a computer science student]

## Course Information

| Item | Details |
| --- | --- |
| Course | CC 101 Introduction to Computing |
| Instructor | [Instructor's name] |
| Academic Year | 2026–2027 |

## Portfolio Contents

1. [Week 7 – Pseudocode and Flowcharts](week07-algorithms/)
2. More activities will be added each week.

## Sample Pseudocode: Decimal to Binary

```
[Paste your pseudocode from Box 1 here]
```

My Workspace

![My laboratory computer](images/my-pc.jpg)

Last updated: [date in YYYY-MM-DD format]
```

15. Save the file. Click **File** → **Save as**. Go to **Documents** → **cc101-portfolio**. Type **README.md** as the **File name**, set **Save as type** to **All files**, make sure **Encoding** is **UTF-8**, and click **Save**.

16. Verify the file. In File Explorer, open the cc101-portfolio folder and confirm the file is named **README.md**, not README.md.txt. If it ends in .txt, right-click it, choose **Rename**, and remove .txt. Then press the **Windows key**, type **cmd**, press **Enter**, and run the commands below. Record the results in **Table 5**.

    cd %USERPROFILE%\Documents\cc101-portfolio

    dir

    type README.md

    Tip: If "The system cannot find the path specified" appears, your Documents folder may be inside OneDrive. Try **cd %USERPROFILE%\OneDrive\Documents\cc101-portfolio** instead.

17. Optional preview: if your Notepad has a formatted view, switch to it from the **View** menu and note in Table 5 which elements appear formatted. Some elements, such as tables, images, and code blocks, may still appear as plain text. GitHub will display them correctly.

18. Copy the whole **cc101-portfolio** folder to your USB flash drive.

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Part F – Peer Review

19. Exchange seats with a classmate. Open their README.md in Notepad and complete the checklist in **Table 6** for their file. Write your name as the reviewer and return the worksheet. Fix any missing element in your own file before leaving.

Part G – Take-Home: Upload to GitHub

Do this part when you have Internet access (at home, on campus Wi-Fi, or on your phone's browser in desktop mode).

20. Go to **github.com** and click **Sign up**. Use an email you check regularly and choose a professional username, such as your name. Verify your email address.
21. Click the **+** button at the upper right and choose **New repository**. Type **cc101-portfolio** as the **Repository name**, choose **Public**, and leave **Add a README file** unchecked. Click **Create repository**.
22. On the setup page, click the link **uploading an existing file**. Drag your README.md (and your week07-algorithms folder, if it has files) onto the page.
23. Under **Commit changes**, type **Add initial README** as the message and click **Commit changes**.
24. Open your repository's main page and check that the README appears formatted. Copy the repository URL (for example, https://github.com/username/cc101-portfolio), record it in **Table 7**, and submit it through the LMS.

    Note: GitHub does not store empty folders. The link to week07-algorithms will work after that folder contains at least one file, and the image will appear after you upload images/my-pc.jpg.

V. Data and Findings

Table 1. Problem Analysis for Problem 1 (Part A)

Item	Your Answer
Restated problem (your own words)	Convert a non-negative decimal number into its equivalent binary number by repeatedly dividing the number by 2 and collecting the remainders.
Input	A non-negative decimal integer, such as 13 or 25.
Output	The equivalent binary representation of the decimal integer.
Processes (operations)	1. Set the binary result to an empty string. 2. Divide the number by 2. 3. Get the remainder using n mod 2. 4. Add or prepend the remainder to the binary result. 5. Replace n with the integer quotient using n \operatorname{DIV} 2. 6. Repeat the process until n = 0. 7. Display the binary result.
Loop condition (when the loop stops)	Continue looping while n > 0. The loop stops when n = 0. If the input is 0, the output is directly set to 0.
Variables needed	decimal_num - stores the original decimal input; n - stores the changing quotient; remainder - stores the remainder after division by 2; binary - stores the binary result.
13 and 25 in binary (computed by hand)	13 = 1101 and 25 = 11001

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Box 1. Pseudocode for Problem 1 (Part B)

```
START
  INPUT decimal_num
  IF decimal_num == 0 THEN
    SET binary = "0"
  ELSE
    SET binary = ""
    SET n = decimal_num
    WHILE n > 0 DO
      SET remainder = n MOD 2
      SET binary = remainder + binary
      SET n = n DIV 2
    ENDWHILE
  ENDIF
  DISPLAY binary
END
```

Figure 1. Flowchart for Problem 1 (Part C)

Draw your flowchart here.

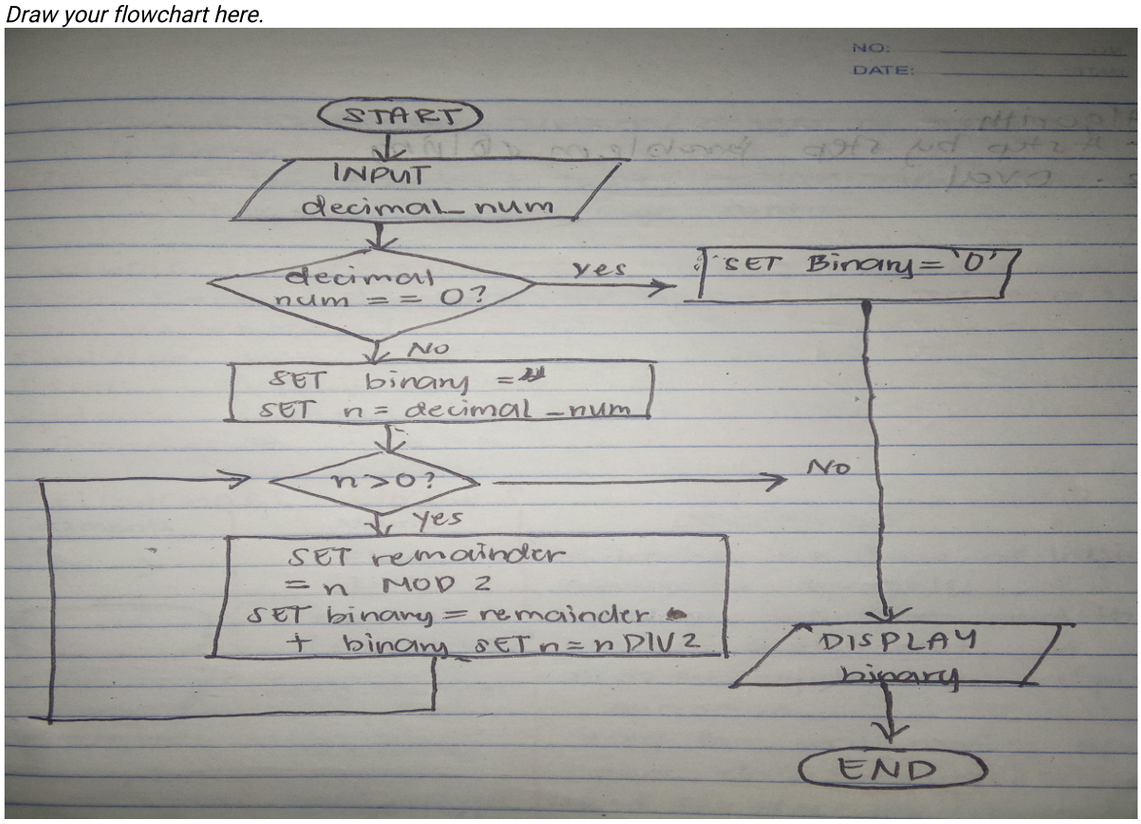


Table 2. Trace of Problem 1 with Input 13 (Part C)

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)
1	13	1	6	1
2	6	0	3	01

Pass	n (before)	n MOD 2 (remainder)	n DIV 2 (new n)	binary (so far)
3	3	1	1	101
4	1	1	0	1101
5	-	-	-	-

Value displayed at the end: __1101_____

Table 3. Test Cases for Problem 1 (Part C)

Case	Decimal Input	Binary (by hand)	Binary From Your Pseudocode	Match? (Y/N)
1	0	0	0	Y
2	5	101	101	Y
3	13	1101	1101	Y
4	25	11001	11001	Y
5	255	11111111	11111111	Y

Box 2. Pseudocode for Problem 2 (Part D)

```
BEGIN
  INPUT binary_string
  SET decimal_value = 0
  SET base = 1
  SET length = length of binary_string

  FOR i FROM (length - 1) DOWNTO 0 DO
    SET current_bit = character at index i of binary_string

    IF current_bit == '1' THEN
      decimal_value = decimal_value + base
    ENDIF

    base = base * 2
  ENDFOR

  DISPLAY decimal_value
END
```


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Figure 2. Flowchart for Problem 2 (Part D)

Draw your flowchart here.

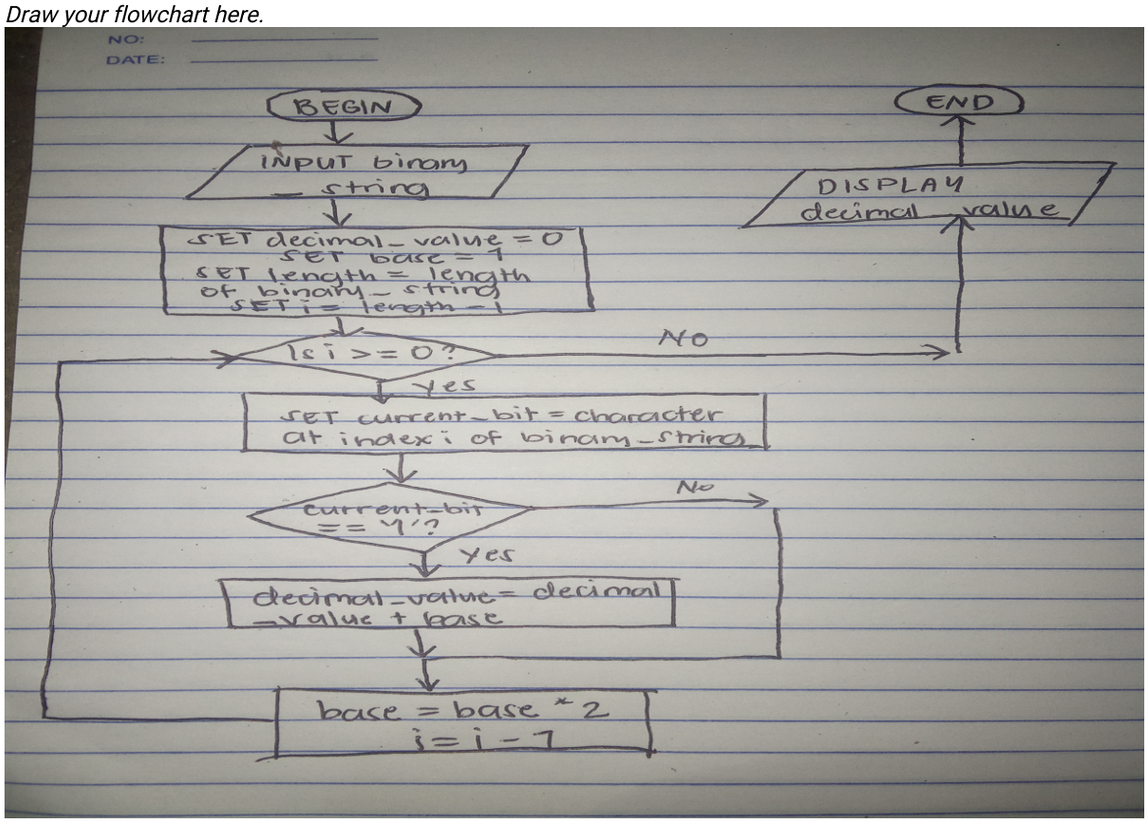


Table 4. Test Cases for Problem 2 (Part D)

Case	Binary Input	Decimal (by hand)	Decimal From Your Pseudocode	Match? (Y/N)
1	1101	13	13	Y
2	101010	42	42	Y
3	11111111	255	255	Y
4	10000000	128	128	Y

Bonus Box. Extended Pseudocode: Decimal to Base 2, 8, or 16 (Optional)

```
START

INPUT decimal_num
INPUT base

IF base is not 2 AND base is not 8 AND base is not 16 THEN
   DISPLAY "Invalid base"
ELSE IF decimal_num == 0 THEN
   DISPLAY "0"
ELSE
   SET result = ""
   SET n = decimal_num

   WHILE n > 0 DO
      SET remainder = n MOD base

      IF remainder == 10 THEN
         SET digit = "A"
      ELSE IF remainder == 11 THEN
         SET digit = "B"
      ELSE IF remainder == 12 THEN
         SET digit = "C"
      ELSE IF remainder == 13 THEN
         SET digit = "D"
      ELSE IF remainder == 14 THEN
         SET digit = "E"
      ELSE IF remainder == 15 THEN
         SET digit = "F"
      ELSE
         SET digit = remainder
      ENDIF

      SET result = digit + result
      SET n = n DIV base
   ENDWHILE

   DISPLAY result
ENDIF

END
```

Table 5. README.md File Check (Part E)

Item	Your Findings
Full folder path of cc101-portfolio	C:\Users\Roxanne\Documents\cc101-portfolio
File name shown in File Explorer	README.md
Type shown in File Explorer (Details view)	MD File
Size of README.md shown by dir (bytes)	1,248 bytes

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Item	Your Findings
Does type README.md show your Markdown text? (Y/N)	Y
Notepad formatted view available? (Y/N)	N
Elements that appeared formatted in Notepad (if available)	None

Table 6. Peer Review Checklist (Part F)

Reviewer's name: _____ Owner of the README: _____

Element	Markdown Used	Present and Correct? (Y/N)
Level-1 heading	#	Y/N
Level-2 headings	##	Y/N
Paragraph	Text with a blank line before and after	Y/N
Bold and italic text	**bold** *italic*	Y/N
Bulleted list	- item	Y/N
Numbered list	1. item	Y/N
Table	| --- | header separator	Y/N
Link	[text](url)	Y/N
Image	![alt](path)	Y/N
Code block	Lines between two lines of three backticks	Y/N
Horizontal rule	---	Y/N
File saved as README.md (not .txt)	_	Y/N

Table 7. GitHub Submission (Part G, Take-Home)

Item	Your Answer
GitHub username	roxannegracela7-ui
Repository URL	https://github.com/roxannegracela7-ui/cc101-Introduction-to-Computing-.git
Date uploaded (YYYY-MM-DD)	2026-10-16
Does the README display formatted on GitHub? (Y/N)	Y
Elements that did not display as expected, and why	Left blank

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VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Which was easier for you to create, the pseudocode or the flowchart? Why?

For me, the pseudocode was easier to make because I could explain the steps using words and simple commands. It was also easier to edit whenever I found something wrong. After making the pseudocode, I used it as a guide when creating the flowchart.

2. Did tracing and testing reveal any errors in your pseudocode or flowcharts? Describe what you found and how you corrected it.

Yes, I found a few errors when I traced and tested my pseudocode and flowchart. Some of the calculations and conditions were not correct, so I checked each step again. I fixed the errors involving MOD, DIV, and the loop condition. After that, I tested the program again to check if the answers were already correct.

3. How is writing a README in Markdown different from typing a document in Microsoft Word?

A README written in Markdown uses symbols and text to create headings, lists, links, and other formats. In Microsoft Word, I can format the document by using buttons and menus. Markdown is more text-based, while Word allows me to see the formatting while I am editing the document.

4. How can a GitHub portfolio help you as a future computing professional?

A GitHub portfolio can help me show my projects and programming skills to my teachers, future employers, or clients. It allows me to organize and share my work online. It can also help me gain experience in documenting projects and build a good portfolio for my future career in computing.

VII. Scoring Guide

Criteria	Description	Points
Problem Analysis & Pseudocode	Table 1 is complete; the pseudocode for both conversions is correct (correct use of MOD, DIV, and the loop), uses proper keywords, and is indented.	25
Flowcharts & Testing	Both flowcharts use the correct symbols with labeled decisions and a clear loop; the trace (Table 2) and test cases (Tables 3 and 4) are correct.	25
README.md	The file is saved correctly as README.md, uses all required Markdown elements, and passes the peer review in Table 6.	25
GitHub Upload	The README is uploaded to a public cc101-portfolio repository and the URL is submitted (Table 7).	10
Realization / Reflection	Answers are thoughtful, specific to the student's own work, and well written.	10
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	5

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Criteria	Description	Points
Total		100

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